UNIVERSITY OF ENGINEERING AND TECHNOLOGY LAHORE



AI-542 Agentic AI

Project 1 Report

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Abstract

This study examines how three decoding parameters—temperature, Top-K, and Top-P—relate to the output behaviour of two locally run large language models, Llama 3 and Qwen 2.5. The experiment used one
creative-writing prompt requesting an imaginative story in exactly 80 words, with four temperature values (0, 0.3, 0.7, 1.0), three Top-K values (10, 40, 100), three Top-P values (0.5, 0.9, 1.0), and two repeats per configuration.
The supplied data contain 144 successful responses across 72 model-configuration groups. Using whitespace-separated word counting, 2 responses (1.39%) contained exactly 80 words. 71 of 72 configurations had identical normalized responses across their two repeats. Textual sensitivity was estimated with Python's SequenceMatcher ratio between one representative response at adjacent parameter settings. Results indicate that exact length compliance was uncommon in this dataset and that some parameter changes coincided with substantial lexical changes. Because the experiment uses one prompt, two repeats, and lexical rather than semantic evaluation, the findings are exploratory and should not be generalized to overall model quality or other task types.

Keywords: *large language models, sampling, temperature, Top-K, Top-P, reproducibility, word-count compliance, text similarity.*

1. Introduction

Large language models (LLMs) generate text by repeatedly selecting a next token from a probability distribution. The decoding strategy affects which tokens remain eligible and how strongly the model favours higher-probability options. Temperature, Top-K, and Top-P are common controls used to alter the diversity and determinism of generated text. Their effects matter in creative generation, where variation may be desirable, and in constrained tasks, where consistency or exact-format compliance may be more important.

This project compares two locally run models, Llama 3 and Qwen 2.5, under a controlled grid of sampling settings. It records repeated outputs, computes repeat-consistency and lexical-similarity measures, and evaluates whether outputs comply with a prompt requiring exactly 80 words. The analysis is designed to be transparent and reproducible using saved CSV results and a Streamlit dashboard.

2. Aim and Research Questions

Aim. To explore how temperature, Top-K, and Top-P settings relate to repeat consistency, lexical variation, and exact word-count compliance for Llama 3 and Qwen 2.5 on a creative-writing prompt.

Research questions

- RQ1: How consistent are repeated outputs under the same sampling configuration?
- RQ2: How much does lexical wording change between adjacent values of temperature, Top-K, and Top-P when other parameters are held fixed?
- RQ3: How often do the models satisfy the prompt's exactly-80-word constraint?
- RQ4: Do the two models show different observed patterns in this experiment?

3. Background

3.1 Temperature

Temperature rescales token logits before sampling. Lower values generally concentrate probability on more likely tokens, while higher values can increase variation. The effect depends on the model, prompt, decoding implementation, and other sampling settings.

3.2 Top-K and Top-P

Top-K limits sampling to a fixed number of high-probability candidate tokens. Top-P (nucleus sampling) retains a set of candidates whose cumulative probability reaches a specified threshold. These methods restrict the candidate set differently and can interact with temperature.

3.3 Evaluation of generated text

Repeated-output agreement and lexical similarity are useful descriptive signals, but they are not equivalent to quality, creativity, semantic similarity, factual correctness, or safety. Exact word-count compliance is a
task-specific measure and is directly relevant to the prompt used here. The counting convention

must be stated because tokenizers and word-count rules can differ.

4. Methodology

4.1 Experimental setup

The experiment was run locally with Ollama and analysed in Python. The project also includes a Streamlit dashboard for exploring the generated responses and parameter sensitivity. The report uses the supplied responses.csv and metrics.csv files as the experimental record.

4.2 Prompt and sampling grid

The single prompt was: *"Write a short, imaginative story in exactly 80 words about a robot discovering a hidden garden on Mars."*

Factor	Values
Model	llama3; qwen2.5
Temperature	0.0, 0.3, 0.7, 1.0
Top-K	10, 40, 100
Top-P	0.5, 0.9, 1.0
Repeats per configuration	2
Planned outputs	2 × 4 × 3 × 3 × 2 = 144
Unique model/configuration groups	2 × 4 × 3 × 3 = 72

4.3 Measures

- **Successful responses:** records with no recorded error and a non-empty response.
- **Exact repeat consistency:** agreement based on normalized response text within a model/prompt/parameter group. Normalization lowercases text and collapses whitespace.
- **Mean pairwise text similarity:** SequenceMatcher ratio computed between normalized repeated responses within a configuration.
- **Parameter sensitivity:** SequenceMatcher ratio between representative responses at adjacent parameter values, holding the other two parameters fixed. The first repeat in sorted order was selected for each setting, matching the dashboard's calculation.
- **Exact 80-word compliance:** a response is compliant if len(response.split()) == 80. This is whitespace-separated counting, not model-token counting.
- **JSON validity and reference similarity:** retained as general pipeline metrics, but not interpreted as meaningful outcome measures for this creative-writing prompt. No reference answer was supplied.

4.4 Analysis procedure

The analysis summarizes response counts, word counts, configuration-level consistency, and adjacent-setting lexical similarity. The sensitivity results are descriptive comparisons rather than

causal estimates. Because the prompt is fixed, the study does not estimate general performance across tasks.

5. Results

5.1 Dataset integrity and response length

The supplied response file contained 144 usable, non-empty responses. The experimental design planned 144 responses. The median response length was 79 words, the mean was 72.78, and observed lengths ranged from 48 to 91 words under whitespace-separated counting.

Model	Responses	Mean words	Median	Min–max	Exactly 80	Compliance
llama3	72	85.61	87	79–91	2	2.78%
qwen2.5	72	59.94	60	48–88	0	0.00%
Overall	144	72.78	79	48–91	2	1.39%

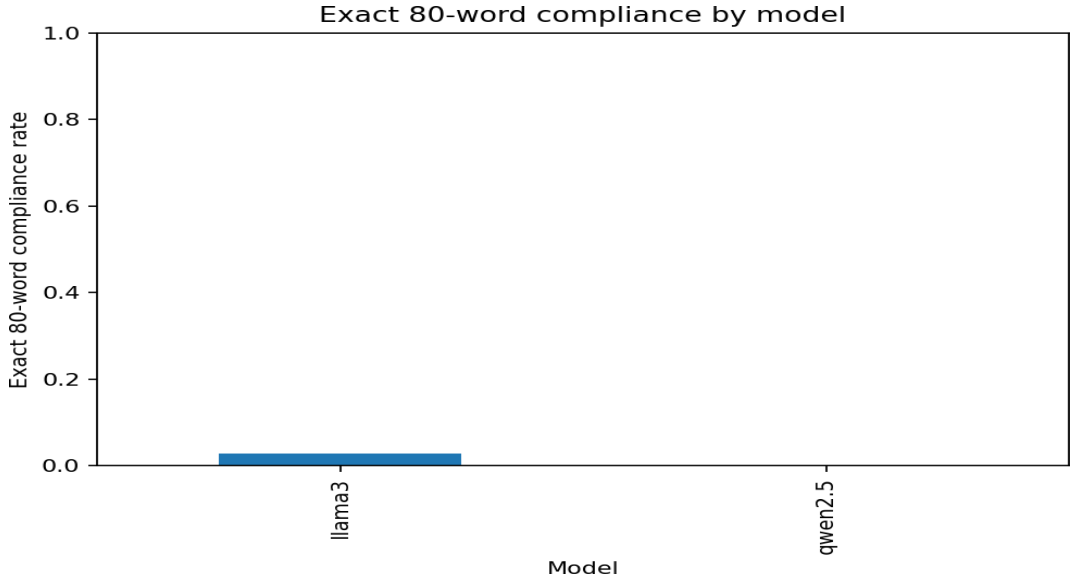


Figure 1. Proportion of responses meeting the exactly-80-word requirement.

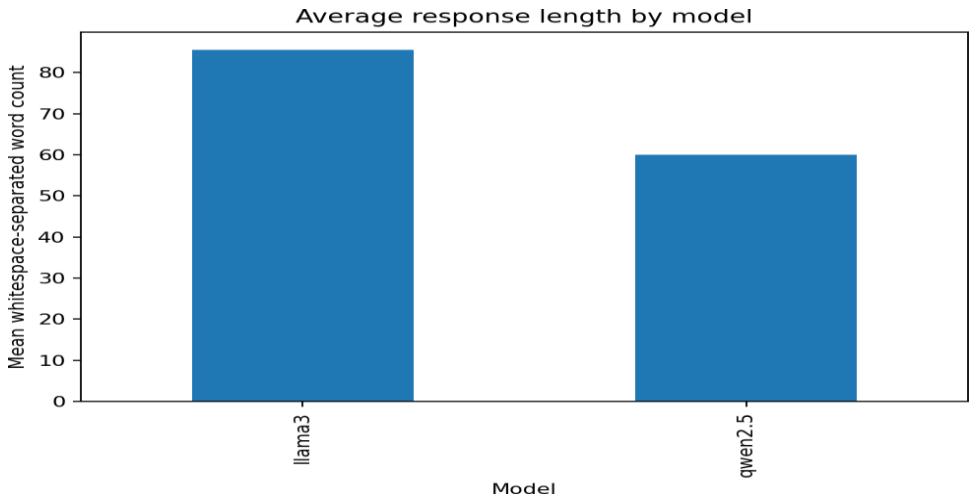


Figure 2. Mean whitespace-separated word count by model.

Only 2 of 144 responses (1.39%) contained exactly 80 whitespace-separated words. This indicates that the models frequently did not meet the explicit length constraint, even though the prompt stated it directly. This result should be interpreted as compliance with this particular counting rule, not as a general measure of writing quality.

5.2 Repeat consistency

At the configuration level, 71 of 72 groups (98.61%) had an exact consistency rate of 1.0. The mean configuration-level exact consistency rate was 98.61%.

Model	Mean exact consistency	Configurations
llama3	100.00%	36
qwen2.5	97.22%	36

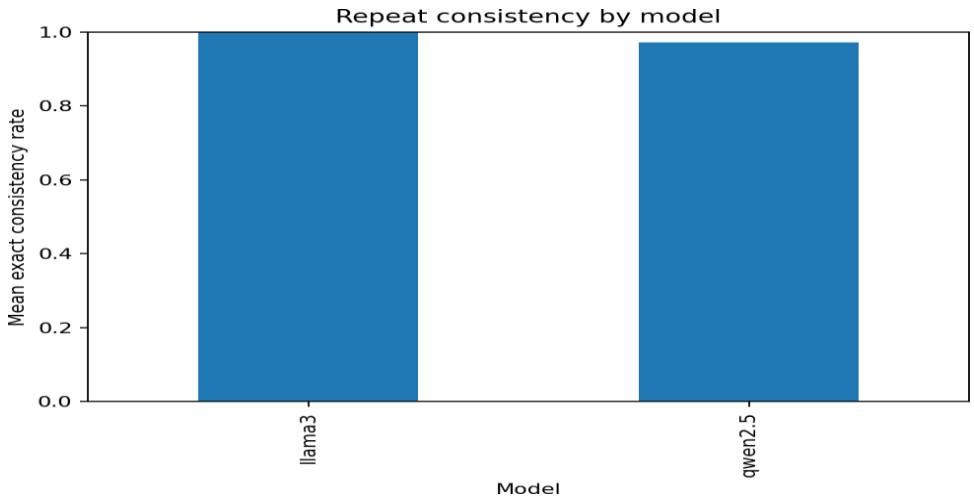


Figure 3. Mean exact repeat consistency by model.

The only configuration with non-identical normalized repeats was qwen2.5 at temperature 0.0, Top-K 10, and Top-P 0.5. Its mean pairwise text similarity was 0.590. This single observation should not be overinterpreted.

5.3 Parameter sensitivity

The following table reports lexical similarity between adjacent settings using the first repeat at each setting. Higher values indicate more similar wording; lower values indicate greater lexical difference. They do not measure semantic similarity or quality.

Model	Parameter	Transition	Similarity
llama3	Temperature	0.0 □ 0.3	0.203
llama3	Temperature	0.3 □ 0.7	0.114
llama3	Temperature	0.7 □ 1.0	0.293
llama3	Top-K	10 □ 40	0.094
llama3	Top-K	40 □ 100	1.000
llama3	Top-P	0.5 □ 0.9	0.099
llama3	Top-P	0.9 □ 1.0	0.250
qwen2.5	Temperature	0.0 □ 0.3	0.213
qwen2.5	Temperature	0.3 □ 0.7	0.055
qwen2.5	Temperature	0.7 □ 1.0	0.025
qwen2.5	Top-K	10 □ 40	0.173
qwen2.5	Top-K	40 □ 100	0.233
qwen2.5	Top-P	0.5 □ 0.9	0.151
qwen2.5	Top-P	0.9 □ 1.0	0.179

Observed examples include a high similarity score of 1.000 for Llama 3 between Top-K 40 and 100 at temperature 0.7 and Top-P 0.9, meaning the selected normalized response strings were identical for that comparison. Other transitions produced lower similarities, showing lexical changes under the selected conditions. These patterns are specific to this prompt and selected fixed settings.

5.4 Other metrics

The JSON-validity rate was 0% in the supplied metrics, but the prompt did not request JSON, so this is not evidence of task failure. Reference lexical similarity was unavailable because no reference answer was supplied. The experiment did not include a validated hallucination or factuality evaluation.

6. Discussion

The most direct result is that exact length compliance was poor: only two responses met the 80-word target under the declared whitespace-counting rule. This illustrates a practical distinction between producing plausible prose and following a precise output constraint. Applications that require exact length should use explicit
post-generation validation and, where necessary, a repair or constrained-generation step.

Repeated outputs were highly consistent in this dataset: 71 of 72 configuration groups had identical normalized responses across their two repeats. This may reflect the decoding setup, model behaviour, or
implementation-specific determinism. Two repeats per configuration are not enough to characterize the full distribution of possible outputs.

Parameter sensitivity varied across transitions. The lexical similarity score can show whether surface wording changed, but it cannot establish that a response became more creative, coherent, accurate, or useful. Similarity may also be affected by response length. A stronger follow-up study would combine automated measures with human ratings or task-specific checks.

The results do not support a broad claim that one model is generally better than the other. They support only descriptive statements about these models, this prompt, this parameter grid, and the supplied local-run outputs.

7. Limitations and Threats to Validity

- **Single prompt:** one creative-writing prompt cannot represent summarization, reasoning, coding, question answering, or other tasks.
- **Small number of repeats:** two repeats per configuration provide limited evidence about variability.
- **Word-count convention:** whitespace splitting is transparent and reproducible but can differ from human editorial conventions and other tokenization rules.
- **Lexical metric limitations:** SequenceMatcher is a character-sequence comparison, not a semantic or quality metric.
- **Configuration interactions:** sensitivity comparisons vary one parameter at a time at selected fixed values; results may differ at other settings.
- **Execution context:** hardware, exact model tags, quantization, Ollama version, runtime options, and random seeds should be recorded for replication if available.
- **No human evaluation:** the project did not include blinded ratings of creativity, coherence, relevance, or readability.

- **No factuality test:** a fictional story prompt and no reference answer do not support a hallucination-rate estimate.

8. Conclusion and Future Work

This project implemented a controlled sampling grid comparing Llama 3 and Qwen 2.5 across temperature, Top-K, and Top-P settings. The analysed dataset contained 144 successful responses and 72 configuration groups. The principal findings are that only 2/144 responses (1.39%) met the exact 80-word requirement under whitespace-based counting, while 71/72 configurations showed identical normalized responses across repeats. Adjacent-setting lexical similarity varied, but it should be interpreted only as a measure of surface-form overlap.

Future work should use multiple prompts across task categories, increase repeats, record model and runtime versions, and report uncertainty. Evaluation could include a word-count validation metric, human ratings for creative quality, semantic similarity where suitable, and task-specific correctness tests. For practical use, exact-format constraints should be validated programmatically rather than assumed to be followed.

References

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Appendix A. Experimental Configuration

Prompt: Write a short, imaginative story in exactly 80 words about a robot discovering a hidden garden on Mars.

Models: llama3 and qwen2.5.

Sampling values: temperature = 0, 0.3, 0.7, 1.0; Top-K = 10, 40, 100; Top-P = 0.5, 0.9, 1.0; repeats = 2.

Planned request count: 144. The supplied CSV files were used as the record of observed outputs; no additional generations were run for this report.

Reproducibility note: Preserve the raw response CSV, metrics CSV, prompt definition, model tags, Ollama version, and code revision alongside any future replication.